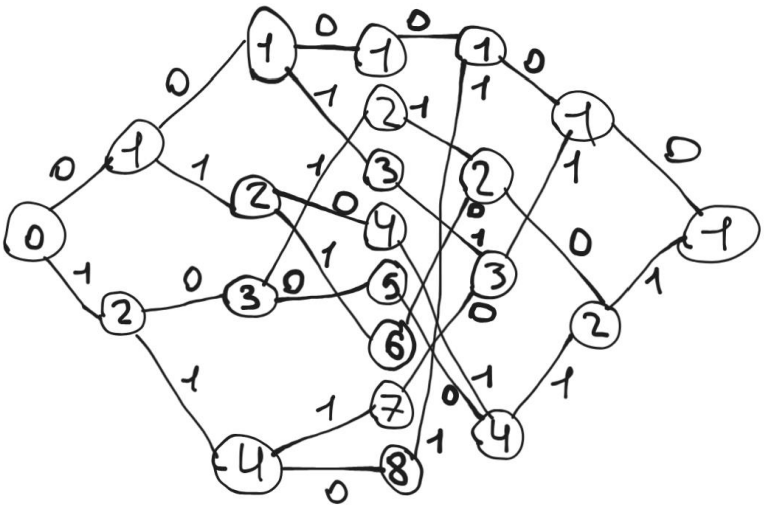
$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

7 ярусов

I) По кодовым словам:

$C_0 = 000000$
$C_1 = 101101$
$C_2 = 001110$
$C_3 = 010111$
$C_4 = 110011$
$C_5 = 011001$
$C_6 = 111010$
$C_7 = 110100$

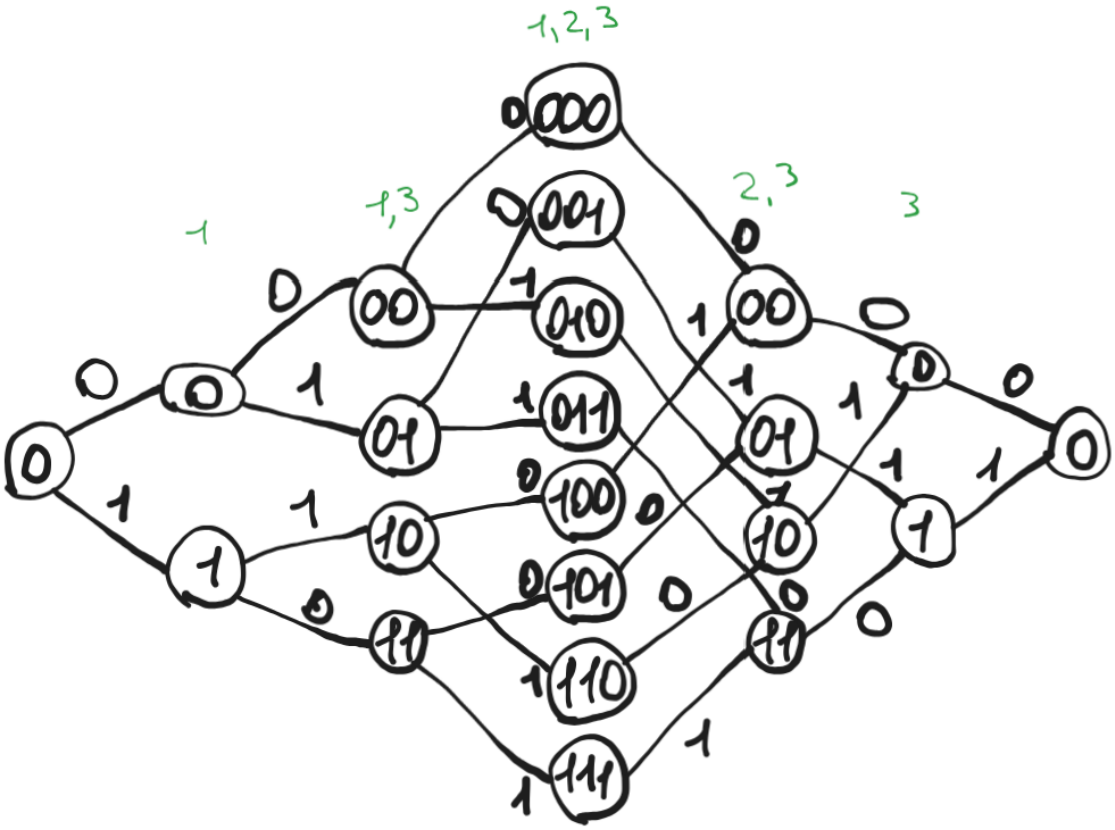
i	$\{C^p\}$	$F_i(C^p)$	$\{V_{i-1}\}$	V_i	C_i
0	$\varnothing$	$\{C_m\}$	$\varnothing$	1	—
1	0	00000, 01110, 10111, 11001	1	1	0
	1	01101, 00011, 11010, 10100	1	2	1
2	00	0000, 1110	1	1	0
	01	0111, 1001	1	2	1
	10	1101, 0011	2	3	0
	11	1010, 0100	2	4	1
3	000	000	1	1	0
	101	101	3	2	1
	001	110	1	3	1
	010	111	2	4	0
	100	011	3	5	0
	011	001	2	6	1
	111	010	4	7	1
	110	100	4	8	0
4	0000	00	1	1	0
	1101	01	2	2	1
	1011	10	3	3	0
	0110	11	4	4	0
	0011	0000	1	1	0
	1110	1110	2	2	1
	0101	1001	4	4	1
	1000	0111	5	5	0
5	00000	0	1	1	
	00111	1	2	2	
	11101		3		
	11010		1		
	10110		2		
	01011		4		
	10001		5		
	01100		2		
6	$\{C_m\}$	$\varnothing$	1,2	1	



II) По порожд. матр.

МСФ:

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} \boxed{110}1 & 0 & 0 \\ 00\boxed{11}1 & 0 \\ 0\boxed{1011}1 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$



III) Синдромная рем.:

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} = [I \;\; P]$$

$$H = (P^T \;\; I) = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

к.сл:

000000
101101
001110
010111
100011
011001
111010
110100

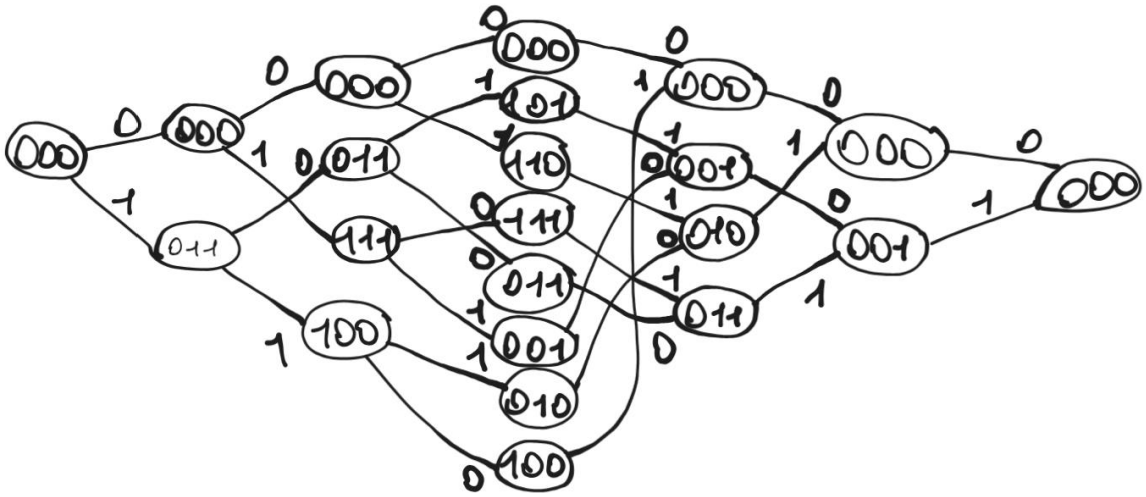
i	синдром
0	000
1	011

00	000
10	011
01	111
11	100

000	000
101	101
001	110
010	111
100	011
011	001
111	010
110	100

000000	000
101101	001
001110	010
010111	011
100011	011
011001	001
111010	010
110100	000

000000	000
101110	001
001111	000
010111	001
100001	001
011100	001
111101	000
110010	000

000000
101101
001110
010111
100011
011001
111010
110100

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